



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering

CSE325: System Analysis and Design

Batch: 23rd

Semester: Spring 2025

Date: 13-09-2025

Final Examination

Time: 2 Hours

Total Marks: 40

Answer any five questions of the followings.

5 × 8 = 40

1. (a) What are the reasons of sampling in data collection? Mention the advantages of sampling. 4
- (b) Explain the five steps in interview preparation. 4

2. (a) Illustrate pyramid structure for interviewing. 3
- (b) The cost of developing a website over a five-month period as follows: 5

At the end of	Stage	Estimated Cost	Cumulative Estimate	Estimated Duration	Stage Completed	Actual Cost of Stage to Date	Actual Cost of Project to date
Month 1	Stage 1	\$6,000	\$6,000	1 month	100%	\$6,000	\$6,000
Month 2	Stage 2	3,000	9,000	1 month	100%	3,000	9,000
Month 3	Stage 3	3,000	12,000	1 month	100%	3,000	12,000
Month 4	Stage 4	3,000	15,000	1 month	50%	5,000	17,000
Month 5	Stage 5	3,000	18,000	1 month	0%	Not yet begun	Not yet begun

If the team is able to complete only 50 percent of stage 4 at the end of the fourth month, then calculate the revised budget Estimate at Completion (EAC).

3. (a) List the benefits of data dictionary. 3
- (b) Create a data dictionary for the registration and login process of a library management system. 5
4. (a) What is the objective of designing user interface? 3
- (b) List ten factors that must be considered when choosing output technology. 5
5. (a) State the rules for constructing a Zero level diagram. 4
- (b) Assume that, the proportion of the population having the attribute $p = 0.05$, acceptable interval estimate $t = \pm 0.02$. Now, calculate the sample size if the confidence level is 90%. 4

Confidence Level (%)	Confidence Coefficient (z value)
99	2.58
98	2.33
97	2.17
96	2.05
95	1.96
90	1.65
80	1.28
50	0.67

6. (a) Explain Electronic Device Repair Management System with E-R diagram. 4
- (b) Draw a physical DFD for Customer Order Process in supermarket. 4
7. The Nokia consultant company wants to update the accounting functions, a maker of digital cameras, you as a system analyst have to interview Mr. Murad, the chief accountant. Write four to six interview objectives covering his use of decision-making frequency, desired qualities of information, and decision-making style. 8
- i. Write down how you will approach Mr. Murad to set up an interview.
 - ii. State which structure you will choose for the interview. Why?
 - iii. Mr. Murad has four subordinates who also use the system. Would you interview them also? Why or why not?
 - iv. Would you also try to interview customers (visitors to the Web site)? Are there any better ways to get the opinions of customers? Why or why not?



HAMDARD UNIVERSITY BANGLADESH

Department of Computer Science and Engineering
Faculty of Science, Engineering and Technology
Semester Final Examination 2025

Course Code: MAT323

Final Examination

Date: 9/10/2025

Total Time: 02:00 Hours

Course Title: Numerical Analysis

Semester: Spring 2025

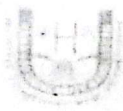
Total Marks: 40

Answer any four (04) out of the following questions:

4 × 10 = 40

[The figures in the right margin indicate full marks. Symbols used have their usual meaning.]

- | No. | Questions | CLOs | Marks | | | | | | | | | | | | | | | | | | |
|-------|---|-------|-------|-----|-----|-----|-----|-------|---|---|-------|-----|-----|-----|-----|-----|-----|-----|-----|--|--|
| Q1. | (a) What is curve fitting? | CLO1 | [1] | | | | | | | | | | | | | | | | | | |
| | (b) Fit a second degree curve to the following data taking x as the independent variable: | CLO3 | [4] | | | | | | | | | | | | | | | | | | |
| | <table border="1"> <tr> <td>x:</td> <td>0</td> <td>1</td> <td>2</td> <td>3</td> <td>4</td> </tr> <tr> <td>y:</td> <td>1</td> <td>5</td> <td>10</td> <td>22</td> <td>38</td> </tr> </table> | x : | 0 | 1 | 2 | 3 | 4 | y : | 1 | 5 | 10 | 22 | 38 | | | | | | | | |
| x : | 0 | 1 | 2 | 3 | 4 | | | | | | | | | | | | | | | | |
| y : | 1 | 5 | 10 | 22 | 38 | | | | | | | | | | | | | | | | |
| | (c) Fit an exponential curve of the form $y = ab^x$ to the following data: | CLO3 | [5] | | | | | | | | | | | | | | | | | | |
| | <table border="1"> <tr> <td>x:</td> <td>1</td> <td>2</td> <td>3</td> <td>4</td> <td>5</td> <td>6</td> <td>7</td> <td>8</td> </tr> <tr> <td>y:</td> <td>1.0</td> <td>1.2</td> <td>1.8</td> <td>2.5</td> <td>3.6</td> <td>4.7</td> <td>6.6</td> <td>9.1</td> </tr> </table> | x : | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | y : | 1.0 | 1.2 | 1.8 | 2.5 | 3.6 | 4.7 | 6.6 | 9.1 | | |
| x : | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | | | | | | | | | | | | | |
| y : | 1.0 | 1.2 | 1.8 | 2.5 | 3.6 | 4.7 | 6.6 | 9.1 | | | | | | | | | | | | | |
| Q2. | (a) Discuss 'Gauss-Seidel' iteration method for solving the system of linear equations. | CLO2 | [4] | | | | | | | | | | | | | | | | | | |
| | (b) Find the solution of the system $83x + 11y - 4z = 95$, $7x + 52y + 13z = 104$, $3x + 8y + 29z = 71$, using Gauss-Seidel method up to three decimals. | CLO3 | [6] | | | | | | | | | | | | | | | | | | |
| Q3. | (a) Define eigen values and eigen vectors. | CLO1 | [2] | | | | | | | | | | | | | | | | | | |
| | (b) What is Euler's method? | CLO1 | [2] | | | | | | | | | | | | | | | | | | |
| | (c) Find the largest eigen values and the corresponding eigen vectors of the following matrix using power method. | CLO3 | [6] | | | | | | | | | | | | | | | | | | |
| | $\begin{bmatrix} 4 & 1 & -1 \\ 2 & 3 & -1 \\ -2 & 1 & 5 \end{bmatrix}$ | | | | | | | | | | | | | | | | | | | | |
| Q4. | (a) What is the initial value problem? | CLO1 | [1] | | | | | | | | | | | | | | | | | | |
| | (b) Derive fourth order Runge-Kutta method for solving first order ordinary differential equation. | CLO2 | [4] | | | | | | | | | | | | | | | | | | |
| | (c) Use Runge-Kutta method to approximate y , when $x = 0.1$ and $x = 0.2$, given that $x = 0$ when $y = 1$ and $\frac{dy}{dx} = x + y$. | CLO3 | [5] | | | | | | | | | | | | | | | | | | |
| Q5. | (a) Solve the following system of linear equations by the Gauss Jordan method.
$2x + 3y - z = 5$, $4x + 4y - 3z = 3$, $-2x + 3y - 2z = 1$. | CLO3 | [4] | | | | | | | | | | | | | | | | | | |
| | (b) Solve the following system of linear equations by the Factorization method.
$x + 5y + z = 21$, $2x + y + 3z = 20$, $3x + y + 4z = 26$ | CLO3 | [6] | | | | | | | | | | | | | | | | | | |
| Q6. | (a) Describe the Picard's method of successive approximations for the solution of a differential equation. | CLO2 | [4] | | | | | | | | | | | | | | | | | | |
| | (b) Using Picard's method, obtain the solution of $\frac{dy}{dx} = x(1 + x^3y)$, $y(0) = 3$. | CLO3 | [3] | | | | | | | | | | | | | | | | | | |
| | (c) Solve by Euler's method of the equation $\frac{dy}{dx} = x + y$, $y(0) = 1$, at the points $x = 0.05$ and $x = 0.10$ taking $h = 0.05$. | CLO3 | [3] | | | | | | | | | | | | | | | | | | |



Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 327: Computer Peripherals & Interfacing

Batch: 23rd

Semester: Spring 2025

Date: 20/09/2025

Final Examination

Time: 2 hours

Total Marks: 40

Answer any **five** questions from the following:

5x8 = 40

- | | | |
|---|--|---|
| 1 | (a) Outline the need of PPI. | 3 |
| | (b) Draw the block diagram of 8155 with the help of 8085. | 5 |
| 2 | (a) Define Memory Mapped I/O. | 2 |
| | (b) Draw the block diagram of 8255 and write about the data bus buffer of 8255 | 6 |
| 3 | (a) Draw the block diagram of Parallel Interface. | 3 |
| | (b) Explain the concept of Small Computer System Interface(SCSI) and draw the possible SCSI system: multiple initiators, multiple targets. | 5 |
| 4 | (a) Explain the functions of the bus management lines in GPIB (IEEE 488). | 4 |
| | (b) Provide an overview of the uses for 7-segment display and draw the table for displaying 26C in 7-segment display. | 4 |
| 5 | (a) Describe the advantages of Stepper motor. | 4 |
| | (b) State the differences between Stepper motor and Servo motor. | 4 |
| 6 | (a) Describe the applications of Stepper motor. | 4 |
| | (b) Explain about MICR, OCR, and OMR. | 4 |
| 7 | (a) Suppose $V_{out} = 6V$, $R_1 = 3.7 \Omega$ and $R_2 = 8 \Omega$. Find out V_{in} for Non-inverting Amp. | 2 |
| | (b) Draw the flow graph of software keyboard interfacing. Write down the principles of raster scan. | 6 |



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 323: Operating System

Batch: 23rd

Semester: Fall 2024

Date: 17/09/2025

Final Examination

Time: 2 Hours

Total Marks: 40

Answer any five questions of the following..

5 × 8 = 40

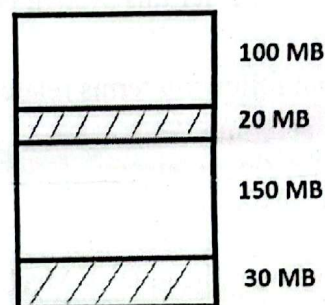
1. (a) Define deadlock and explain the four necessary conditions that must be satisfied simultaneously for a deadlock to occur. 4
(b) Consider the data where Processes: P1, P2, P3, Resources: R1, R2
Allocations: P1 → R1 (Allocated), P2 → R2 (Allocated), P1 → R2 (Requested), P2 → R1 (Requested).
Determine whether this system is in a deadlock. Justify your answer. 4
2. (a) Define the following terms related to disk performance: 4
 - i. Seek time
 - ii. Rotational latency
 - iii. Transfer rate
 - iv. Average access time.(b) Explain Moving-head Disk Mechanism with figure 4
3. (a) What is Shell Script? Why do we need Shell Script? 3
(b) Define how the followings are defined in shell script. 5
 - i. If-Else
 - ii. Case
 - iii. For Loop
 - iv. Shell Arithmetic
 - v. Variables
4. (a) What is a safe state in the context of deadlock avoidance? 2
(b) When a process requests a resource, what condition must the system check before granting the request in the resource allocation graph method of deadlock avoidance? 3
(c) Distinguish between a safe state, an unsafe state, and a deadlock state with the help of a diagram. 3

5. (a) Explain the purpose of the Banker's Algorithm and describe the data structures required for its implementation. 3
- (b) Given the following system state: Determine the safe sequence using Banker's Algorithm 5

Allocation	Max	Available
A B C	A B C	A B C
P0: 0 1 0	7 5 3	3 3 2
P1: 2 0 0	3 2 2	
P2: 3 0 2	9 0 2	
P3: 2 1 1	2 2 2	
P4: 0 0 2	4 3 3	

6. (a) What are Contiguous and Non-Contiguous Memory Allocation? 3
- (b) Apply First-fit, Next-fit, Best-fit, Worst-fit for the following example and Justify which Algorithms is better for variable size partitioning. 5

Process ID	Size
P1	10 MB / 50
P2	15 MB / 75
P3	20 MB / 100
P4	7 MB / 25
P5	8 MB



Primary Memory Partition

7. (a) Explain Hardware Architecture of Paging with figure. 3
- (b) Solve the followings mathematical problems: 5
- If Logical Address is 24bits, find out the Logical Address Space.
 - A process size is 8MB, Physical Address Space is 16MB. If the Page Size is 4KB then calculate Total no of Page and Total no of Frames from the given information.
 - Virtual Address Space is 4GB and Physical Address is 32bits. If the Page Size is 23 bits then calculate Total no of Page and Total no of Frames from the given information.

$1KB = 2^{10}$ $1MB = 2^{20}$ $1GB = 2^{30}$
 $1TB = 2^{40}$



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 321: Theory of Computing

Batch: 23rd

Semester: Spring 2025

Time: 2 Hours

Date: 08/09/2025

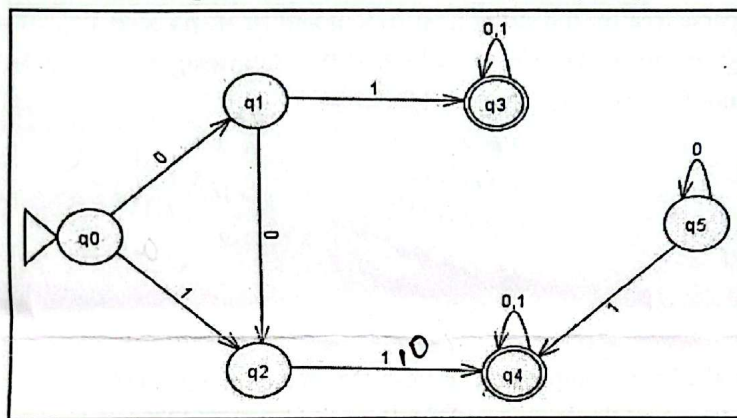
Final Examination

Total Marks: 40

Answer five questions of the following.

5 × 8 = 40

- 1(a). Consider the language $L = \{w | w \text{ contains at least three 1s}\}$. Write a regular language for the given formal language L. 2
- 1(b). Consider the following Deterministic Finite Automata (DFA): 6



Convert the given DFA to minimized DFA.

- 2(a). Define the formal definition of Context-Free Grammar (CFG). 2
- 2(b). Answer each part for the following context-free grammar G: 6 × 1 = 6

$R \rightarrow XRX \mid S$
 $S \rightarrow aTb \mid bTa$
 $T \rightarrow XTX \mid X \mid \epsilon$
 $X \rightarrow a \mid b$

- a. What are the variables of G?
b. What are the terminals of G?
c. Which is the start variable of G?
d. Find three strings in $L(G)$.
e. Find three strings not in $L(G)$.
f. True or False; $T \Rightarrow aba$.

- 3(a). Design a Moore machine to generate 1's complement of a given binary number. 4

3(b). Design a Mealy machine that counts the occurrence of the sequence "abb" in any input string over {a,b}. 4

4(a). What are the rules for a Context-Free Grammar to be in Chomsky Normal Form? 2

4(b). Consider the following grammar : 6

$S \rightarrow ASB$
 $A \rightarrow aAS \mid a \mid \epsilon$
 $B \rightarrow SbS \mid A \mid bb$

Convert this grammar to an equivalent Chomsky normal form.

5(a). Consider a grammar G: 3

$\langle \text{EXPR} \rangle \rightarrow \langle \text{EXPR} \rangle + \langle \text{TERM} \rangle \mid \langle \text{TERM} \rangle$
 $\langle \text{TERM} \rangle \rightarrow \langle \text{TERM} \rangle \times \langle \text{FACTOR} \rangle \mid \langle \text{FACTOR} \rangle$
 $\langle \text{FACTOR} \rangle \rightarrow (\langle \text{EXPR} \rangle) \mid a$

Construct the parse tree for the string $a + a \times a$ and $(a + a) \times a$.

5(b). Define Ambiguity in CFG. Check whether the following grammar is ambiguous or not for string $w = 'aabbccdd'$:

$S \rightarrow AB / C$
 $A \rightarrow aAb / ab$
 $B \rightarrow cBd / cd$
 $C \rightarrow aCd / aDd$
 $D \rightarrow bDc / bc$

Handwritten derivation for $S \rightarrow AB$:
 $S \rightarrow AB$
 $\rightarrow aAbB$
 $\rightarrow aabbB$
 $\rightarrow aabbcbB$
 $\rightarrow aabbccdd$

Handwritten derivation for $S \rightarrow C$:
 $S \rightarrow C$
 $\rightarrow aCd$
 $\rightarrow aDdd$
 $\rightarrow aabDcdd$
 $\rightarrow aabbccdd$

6(a). Briefly explain the basic components of Push Down Automata (PDA). 3

6(b). Design a PDA to accept the language $L = \{a^i b^j c^k \mid i, j, k \geq 0 \text{ and } i = j \text{ or } i = k\}$. 5

7(a). Write the formal definition of Turing Machine. 3

7(b). Design a Turing machine that takes two numbers as input N_1 and N_2 of equal length in binary and computes the logical OR of the two numbers. The tape initially contains $N_1 \# N_2 \#$, where '#' is a tape symbol that is used as the separator. 5

Operating System Lab (Final Exam)

Time: 50 Minutes

Course Title: Operating System Lab

Course Code: CSE 324

Question 1 (10 marks)

Write a shell script that will take n numbers as input from the user and then:

1. Calculate the sum of all numbers.
2. Calculate the product of all numbers.
3. Find the largest and smallest number among them.
4. Calculate the average of all numbers.

Finally, display all the results in a clear format.

Hint: You may use a for loop to read numbers one by one and perform calculations.

Question 2 (10 marks)

Write a shell script to check whether a given number is:

- Positive
- Negative
- Zero

Print a suitable message for each case.

Question 3 (10 marks)

Write a shell script using a for loop to print the multiplication table of a number entered by the user.

Example: If user enters 5, print the table from 5×1 up to 5×10 .

Question 4 (10 marks)

Menu-driven Calculator

Write a shell script that shows the following menu to the user:

1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit

Take two numbers as input.

Ask the user to choose an option (1–5). Use a case statement to perform the chosen operation.

Display the result. Exit when the user chooses option 5.